

Customer Churn Prediction & LTV Engine

Week 4 • Day 7

Final Presentation & Project Handover

Deliver your completed project, final demo, and celebrate 28 days of work

■ Day Objectives

- ◆ Complete the final end-to-end project demo
- ◆ Verify all 28 daily GitHub commits exist
- ◆ Ensure all deliverables are complete and working
- ◆ Write the final project handover document
- ◆ Reflect on learnings and plan next steps

■ 1. Final Project Demo Checklist

Run through this complete demo flow to verify everything works end-to-end:

- PostgreSQL running — customers_clean and customers_ltv tables populated
- python src/db/connection.py — shows 7,043 rows
- python src/features/engineering.py — 15 features created
- models/production_model.pkl, scaler.pkl, feature_columns.pkl all exist
- uvicorn src.api.main:app — starts without errors
- GET http://localhost:8000/health — returns {status: healthy}
- POST http://localhost:8000/predict — returns prediction JSON
- http://localhost:8000/docs — Swagger UI loads

- `pytest tests/test_api.py -v` — all 5 tests pass
- Metabase dashboard loads at `http://localhost:3000`
- `docker-compose up` — both containers start

■ 2. Deliverables Verification

- GitHub repo has 28+ commits, one per day
- `reports/figures/` has 16+ chart images
- `data/processed/` has `telco_features.csv` and `telco_ltv.csv`
- `models/` has: `production_model.pkl`, `ltv_regression.pkl`, `scaler.pkl`, `feature_columns.pkl`, `optimal_threshold.pkl`, `shap_explainer.pkl`
- `README.md` is comprehensive with results table
- Docker: `Dockerfile` and `docker-compose.yml` work
- API documentation complete
- All Python files have docstrings

■ 3. Project Summary Statistics

Fill in your actual results:

■ Final Project Metrics

```
Dataset: 7,043 customers, 21 features
Engineered Features: ~15 additional features
Model: XGBoost AUC-ROC: [YOUR RESULT] ~0.84-0.87
Optimal Threshold: [YOUR RESULT]
Month-to-Month Churn Rate: 42.7%
Charts Generated: 16+
API Endpoints: 5 (/health, /, /predict, /predict/batch, /predict/csv)
API Tests: 5 passing
Daily Commits: 28
Total Development Hours: ~140-180 hours
```

■ 4. What You Have Built — Skills Acquired

- **Data Engineering:** PostgreSQL, SQLAlchemy, ETL pipelines, data warehousing
- **Data Analysis:** Pandas, NumPy, EDA, statistical analysis
- **Data Visualisation:** Seaborn, Matplotlib, Metabase dashboards
- **Machine Learning:** Logistic Regression, Random Forest, XGBoost, cross-validation, feature engineering, handling class imbalance
- **ML Explainability:** SHAP values, feature importance
- **API Development:** FastAPI, REST APIs, Pydantic validation, async endpoints
- **DevOps:** Docker, Docker Compose, containerisation
- **Software Engineering:** Git, GitHub, project structure, testing, documentation

■ 5. Next Steps After This Project

- Deploy to cloud (AWS EC2, Google Cloud Run, or Railway.app)
- Add authentication (OAuth2 / API keys) to the FastAPI
- Implement MLflow for experiment tracking and model versioning
- Set up CI/CD pipeline (GitHub Actions) for automated testing
- Build a Streamlit frontend for the prediction API
- Start Project 2: Real-Time Financial Fraud Detection Pipeline

■ 6. Final Commit & Celebration

■ Final Day 28 Commit

```
git add .
git commit -m "Week4-Day7: Project complete – all 28 days, full stack ML system delivered"
git push origin main

# Verify all commits
git log --oneline | head -30
# Should show 28+ commits

# Create final release
git tag -a v1.0.0-final -m 'Complete 28-day churn prediction project'
git push origin v1.0.0-final
```

■ Estimated Time: 3 – 4 hours

■ **Tip:** Congratulations! You have built a production-level data analytics project from scratch in 28 days. You now have a portfolio project that demonstrates the complete data science lifecycle. Well done!

— End of Week 4 Day 7 — Zaalima Development Confidential —